Mechanics I

– Newton's Laws and Motion –

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Outline

1. Validity of Newton's Laws

2. Newton's First Law: Time, Distance and Inertial Frame

3. Vector Analysis: Coordinate and Vector

4. Newton's Second and Third Laws: Mass and Force

Scientific Method

1. Observe
   - » Observe natural phenomena and observe what happens in the laboratory and accumulate data.
2. Hypothesis
   - » In order to explain the natural phenomena, you can suggest proper theory. We call such theory as Hypothesis.
3. Experiment
   - » Designing and conducting experiments to test the hypothesis.
   - » Collecting and analyzing data from the experiments.
4. Conclusion
   - » Drawing conclusions based on the data and evaluating the hypothesis.
   - » Refining the hypothesis and conducting further experiments to test it.

Laws of Nature

$$\text{hypotheses} \xrightarrow[\text{repetition of experimental verification}]{} \text{laws of nature}$$

- Laws of nature
  - » Newton's laws

$$\mathbf{F} = m\mathbf{a}$$

  - » Maxwell's equations

$$\nabla \cdot \mathbf{E} = 4\pi\rho, \qquad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} + \frac{1}{c}\frac{\partial \mathbf{B}}{\partial t} = 0 \qquad \nabla \times \mathbf{B} - \frac{1}{c}\frac{\partial \mathbf{E}}{\partial t} = \frac{4\pi}{c}\mathbf{j}$$

  - » Copenhagen interpretation

$$i\hbar\frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2\nabla^2}{2m} + V(\mathbf{r})\right]\psi$$

- what if some theoretical/experimental results contradict well-known laws?
  1. you are wrong: oversimplified assumptions, significant neglected effects
  2. you are right: the known laws work only in restricted conditions.
- no laws are claimed to be of universal validity.
  - » the laws have been adequately tested for some classes of phenomena.

Validity and Importance of Classical Mechanics

SCALE	$v \ll c$	$v \lesssim c$
macroscopic	classical mechanics	relativity
microscopic	quantum mechanics	relativistic quantum theory

Table: Classification of Mechanics Based on Velocity and Scale

- classical mechanics is valid in appropriate limiting cases.
- classical mechanics is remarkably successful theory.
  - coherence
  - planetary orbits
  - motions of mechanical operations
  - semi-classical motion of electrons in conductors
- conservation laws of energy, momentum, and angular momentum, discovered in classical mechanics, are of universal validity.

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Time

- Newtonian time
  - » Time exists and is not considered to be nonexistent.
  - » Time is a fundamental and independent dimension used to measure durations and intervals between events.
- what really matters is not how we *define* time, but how we measure it.
  — use clocks = periodic oscillators
- how can we tell if something is in periodic motion?
  - » By going round and round endlessly, we come up with the circular reasoning.
  - » In fact, we accept it and say that we can measure time.

Space

- space can be represented by a coordinate system with an arbitrary reference point.
  - » With knowledge of time and coordinates, we can describe when and where an event occurred.
- how to measure distance? — use some rules and similar others.
  - » triangulation
  - » parallax
  - » electromagnetic wave

Classical Assumption

- we assume that
  1. space and time are continuous.
  2. it is meaningful to say that an event occurred at a specific point in space and instant of time.
  3. there are universal standards of length and time.
     $\longrightarrow$ there is as yet no compelling evidence that we have reached the limits of their range of validity. But. . .

  4. space and time can be measure in any precision.
     $\longrightarrow$ perfectly precise and simultaneous measurement of distances and/or times is not permitted by the Nature itself.

- quantum mechanics

Heisenberg's uncertainty principle

The inequality relating the standard deviation of position σ_x and the standard deviation of momentum σ_p and similarly the energy and time is given by

$$\sigma_x \sigma_p \geq \frac{\hbar}{2} \qquad \sigma_E \sigma_t \geq \frac{\hbar}{2}$$

Classical Assumption

- the last assumption

  5. there is universal time scale. two observers who have synchronized their clocks will always agree about the time of any event: absolute time
     $\longrightarrow$ in actuality, that is not the case within the realm of reality.

  » within the realm of reality, the concept of absolute time is inexistent, instead giving way to the relativity of simultaneity, a phenomenon which has been proven through the special relativity.

Einstein's special relativity

- relativity of simultaneity: time has no absolute meaning. observers can disagree on the simultaneity of events.

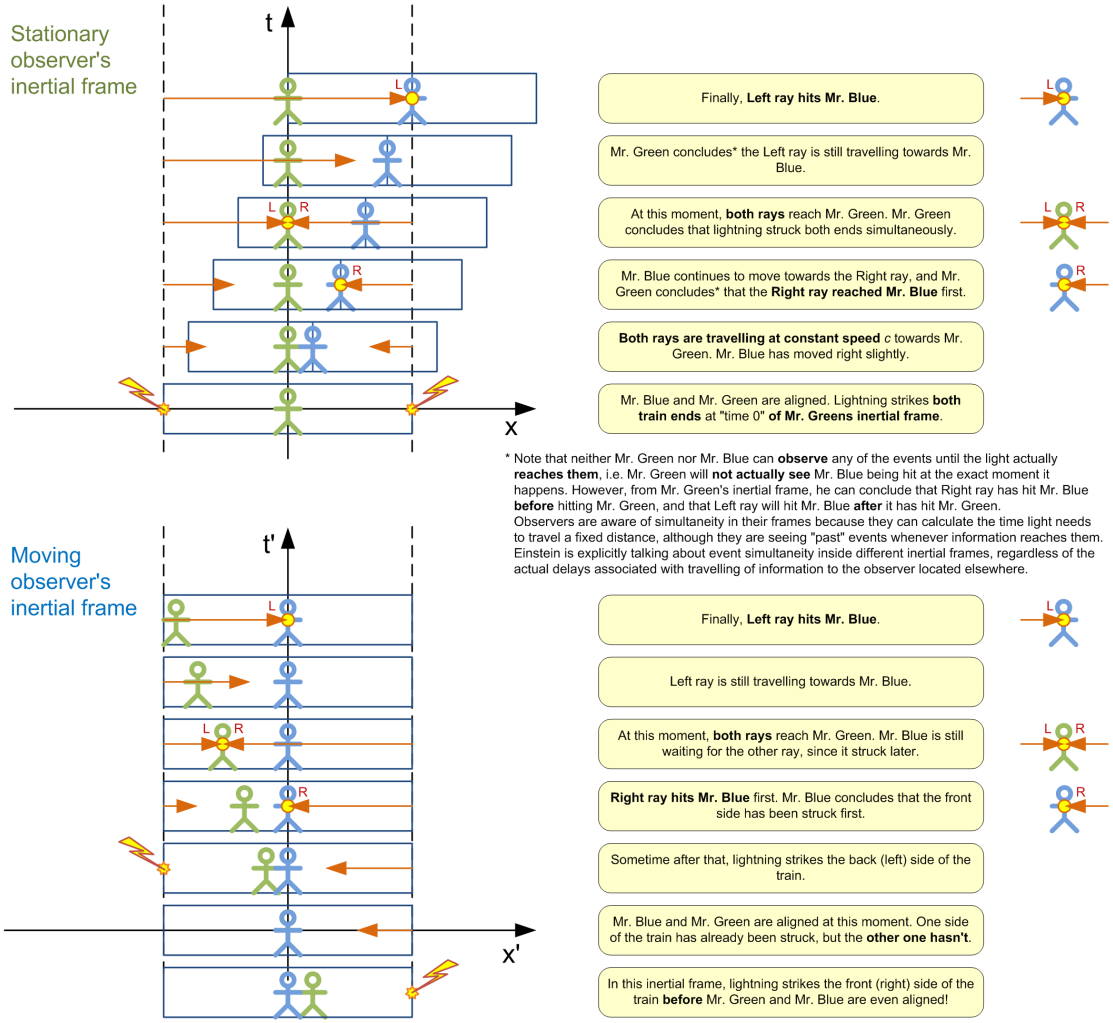
Classical Assumption

- Einstein's special relativity — relativity of simultaneity

Einstein's train thought experiment (1917.)

Comparison between stationary observer's (Mr. Green) and moving observer's (Mr. Blue) inertial frames

NOTE: Diagrams are **not** drawn in scale, they are simply schematics to demonstrate relativity of simultaneity according to Special Relativity theory.



* Note that neither Mr. Green nor Mr. Blue can **observe** any of the events until the light actually **reaches them**, i.e. Mr. Green will **not actually see** Mr. Blue being hit at the exact moment it happens. However, from Mr. Green's inertial frame, he can conclude that Right ray has hit Mr. Blue **before** hitting Mr. Green, and that Left ray will hit Mr. Blue **after** it has hit Mr. Green.
Observers are aware of simultaneity in their frames because they can calculate the time light needs to travel a fixed distance, although they are seeing "past" events whenever information reaches them. Einstein is explicitly talking about event simultaneity inside different inertial frames, regardless of the actual delays associated with travelling of information to the observer located elsewhere.

Relativity Principle

Newtonian mechanics

- the bodies fall downward because they are attracted toward the other object (for example, the Earth), rather than towards some absolute fixed point in space.
- the position and velocity of one body has a meaning only relative to that of the other body.
- Galileo's principle of relativity (classical)
  - » given two bodies moving with constant relative velocity, it is impossible in principle to decide which of them is at rest, and which moving.
  - » all non-accelerating observers are equivalent.
- acceleration retains an absolute meaning
  - » it is experimentally possible to distinguish between motion with uniform velocity (constant in magnitude and direction) and accelerated motion.
  - » unlike an unaccelerating observer, an accelerating observer experiences an inertial force in their reference frame.

Relativity Principle

$\longrightarrow$ above is the principle of relativity in classical mechanics

» all non-accelerating observers are equivalent.

» but, that is not the case in reality and it is proven by the theory of general relativity.

Einstein's general relativity

- in Einstein's theory of general relativity, even acceleration is relative, at least on a small scale, due to the equivalence principle — the effects of being accelerated and of being in a gravitational field are indistinguishable.
- Likewise in the theory of relativity, all unaccelerated observers are equivalent — if two unaccelerated observers perform the same experiment, they must arrive at the same results.

Reference Frame and Inertial Frame

- reference frame — specifying positions and time
  - » zero of time scale
  - » origin in space
  - » set of three (maybe Cartesian) coordinate axes

  $\Longrightarrow$ now we can specify an event $\mathbf{A}(x, y, z, t)$
- reference frames used by different unaccelerated observers are completely equivalent, since the laws of physics expressed in terms of x, y, z, t must be identical with those in terms of x', y', z', t'

However, the laws of physics expressed in terms of the coordinates used by an accelerated observer are different! $\longrightarrow$ there exist the inertial force.

Reference Frame and Inertial Frame

- two kinds of reference frames
  1. inertial frame: reference frames used by unaccelerated observers
  2. non-inertial frame: reference frames used by accelerated observers
- does inertial frame exist?
  - » the Earth is not inertial frame.
  - » idealized definition: an inertial frame, with respect to which any isolate body (far removed from all other matter) move with uniform velocity

Newton's first law

= a body acted on by no forces moves with uniform velocity in a straight line

= in a non-inertial reference frame, it is possible to undergo accelerated motion without external forces.

= "inertial frames" exist

- sometimes it is convenient to use a non-inertial (in particular, rotating) frame

NL-1: Consider a transformation to a relatively uniformly moving frame of reference, where each position vector $\mathbf{r}_i$ is replace by $\mathbf{r}_i' = \mathbf{r}_i - \mathbf{v}t$. Here $\mathbf{v}$ is a constant, the relative velocity of the two frames. How does a relative position vector $\mathbf{r}_{ij}$ transform? How do momenta and forces transform? Show that the equations of motion in the original frame hold in the new one.

Reference Frame and Inertial Frame

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The relative position $\mathbf{r}_{ij}'$ in the new frame is given by

$$\mathbf{r}_{ij}' = \mathbf{r}_i' - \mathbf{r}_j' = (\mathbf{r}_i - \mathbf{v}t) - (\mathbf{r}_j - \mathbf{v}t) = \mathbf{r}_i - \mathbf{r}_j = \mathbf{r}_{ij}$$

That is, the relative position does not change under the transformation. Now consider the velocity in the new frame:

$$\mathbf{v}_i' = \frac{d\mathbf{r}_i'}{dt} = \frac{d\mathbf{r}_i}{dt} - \mathbf{v} = \mathbf{v}_i - \mathbf{v}.$$

Hence, accordingly, the momentum is shifted in the new frame

$$\mathbf{p}_i' = m_i\mathbf{v}_i' = m_i(\mathbf{v}_i - \mathbf{v}) = \mathbf{p}_i = m_i\mathbf{v}.$$

However, the acceleration does not change:

$$\mathbf{a}_i' = \frac{d^2\mathbf{r}_i'}{dt^2} = \frac{d\mathbf{v}_i'}{dt} = \frac{d\mathbf{v}_i}{dt} - \underbrace{\frac{d\mathbf{v}}{dt}}_{=0} = \mathbf{a}_i$$

which means that the equations of motion remain same in the new frame:

$$\mathbf{F}_i = m_i\mathbf{a}_i = m_i\mathbf{a}_i' = \mathbf{F}_i'.$$

The invariance of the forces can be inferred from the fact that the forces are functions of the relative positions and the relative velocities and they do not change:

$$\mathbf{v}_{ij}' = \mathbf{v}_i' - \mathbf{v}_j' = (\mathbf{v}_i - \mathbf{v}) - (\mathbf{v}_j - \mathbf{v}) = \mathbf{v}_i - \mathbf{v}_j = \mathbf{v}_{ij}.$$

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Coordinate Systems

- coordinate systems
  - » a system that uses one or more numbers, referred as coordinates, to uniquely determine the position of the points or other geometric elements on a manifold such as Euclidean space.
- what is coordinates?
  - » let's say that there exist a vector space V over the field K that is a set of objects which can be added and multiplied by elements of K.
  - » then let $V = K^n$ be the set of n-tuples of elements of K. let

$$\mathbf{A} = (a_1, \ldots, a_n) \quad \text{and} \quad \mathbf{B} = (b_1, \ldots, b_n)$$

    be elements of K^n. we call $a_1, \ldots, a_n$ the components, or coordinates of **A**.
  - » we define where $c \in K$

$$\mathbf{A} + \mathbf{B} = (a_1 + b_1, \ldots, a_n + b_n) \quad \text{and} \quad c\mathbf{A} = (ca_1, \ldots, ca_n)$$

  - » when we write K_n, it will always be understood that it is meant as a vector space over K. element of K_n also be called vectors(**A**, **B**) and it is also customary to call elements of an arbitrary vector space vectors.

Coordinate Systems

Definition (Field)

Let K be a subset of the complex numbers $\mathbf{C}$. we shall say that K is a field if it satisfies the following conditions:

1. if x, y are elements of K, then $x + y$ and xy are also elements of K.
2. if $x \in K$, then $-x$ is also an element of K. if furthermore $x \neq 0$, then x^{-1} is an element of K.
3. the elements 0 and 1 are element of K.
4. commutative and associative for addition and multiplication.

Definition (Vector Space)

a vector space V over the field K that is a set of objects which can be added and multiplied by elements of K, in such a way that the sum of two elements of V is again an element of V, the product of an element of V by an element of K is an element of V, and the following properties are satisfied:

1. given elements u, v, w of V, we have

$$(u + v) + w = u + (v + w).$$

Definition (Vector Space)

2. there is an element of V, denoted by O, such that

$$O + u = u + O = u$$

   for all elements u of V.
3. given an element u of V, there exists an element $-u$ in V such that

$$u + (-u) = O.$$

4. for all elements u, v of V, we have

$$u + v = v + u.$$

5. if c is a number, then $c(u + v) = cu + cv$.
6. if a, b are two numbers, then $(a + b)v = av + bv$.
7. if a, b are two numbers, then $(ab)v = a(bv)$.
8. for all elements u of V, we have $1 \cdot u = u$ (1 here is the number one).

Coordinate Systems

- linear combination
  - » let V be an arbitrary vector space, and let $\mathbf{v}_1, \cdots, \mathbf{v}_n$ be elements of V. let $x_1, \cdots, x_n$ be numbers. an expression of type

$$x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n = x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{1m} \end{pmatrix}_1 + \cdots + x_n \begin{pmatrix} a_{n1} \\ \vdots \\ a_{nm} \end{pmatrix}_n$$

- scalar product
  - » let $V = K^n$. let $\mathbf{A}$ and $\mathbf{B} \in K^n$, $\mathbf{A} = (a_1, \cdots, a_n)$ and $\mathbf{B} = (b_1, \cdots, b_n)$. we define the dot product or scalar product

$$\mathbf{A} \cdot \mathbf{B} = a_1 b_1 + \cdots + a_n b_n.$$

- orthogonality
  - » a dot product $\mathbf{A} \cdot \mathbf{B}$ may very well be equal to 0 without either $\mathbf{A}$ or $\mathbf{B}$ being the zero vector. for instance, let $\mathbf{A} = (1, 2, 3)$ and $\mathbf{B} = (2, 1, -\frac{4}{3})$. then $\mathbf{A} \cdot \mathbf{B} = 0$.
  - » we define two vectors $\mathbf{A}, \mathbf{B}$ to be orthogonal (or as well shall also say, perpendicular) if $\mathbf{A} \cdot \mathbf{B} = 0$

Coordinate Systems

- linearly dependence
  - » let V be a vector space over the field K, and let $\mathbf{v}_1, \cdots, \mathbf{v}_n$ be element of V. we shall say that $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly dependent over K if there exist elements $a_1, \cdots, a_n$ in K not all equal to 0 such that

$$a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = O.$$

  - » if there do not exist such numbers, then we say that $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly independent. in other words, vectors $\mathbf{v}_1, \cdots, \mathbf{v}_n$ are linearly independent if and only if the following condition is satisfied:
  - » whenever $a_1, \cdots, a_n$. are numbers such that

$$a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = O.$$

    then $a_i = 0$ for all $i = 1, \ldots, n$.

- basis
  - » if elements $\mathbf{v}_1, \cdots, \mathbf{v}_n$ of V generate (= span) V and in addition are linearly independent, then $\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$ is called a basis of V. we shall also say that the elements $\mathbf{v}_1, \cdots, \mathbf{v}_n$ constitute of form of a basis of V.

Coordinate Systems

- coordinates
  - » let V be a vector space, and let $\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$ be a basis of V. the elements of V can be represented by n-tuples relative to this basis, as follows. if an element v of V is written as a linear combination

$$v = x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n$$

    then by the above remark, the n-tuples $(x_1, \cdots, x_n)$ is uniquely determined by v. we call $(x_1, \cdots, x_n)$ the coordinates of v with respect to our basis, and we call x_i the i-th coordinate. and we say the n-tuple $X = (x_1, \cdots, x_n)$ is the coordinate vector of v with respect to the basis $\{\mathbf{v}_1, \cdots, \mathbf{v}_n\} \xrightarrow[\text{orthogonalize}]{} \underline{\{\mathbf{e}_1, \cdots, \mathbf{e}_n\}}$.

- dimension
  - » let V be a vector space having a basis consisting of n elements. we shall say that n is the dimension of V.

NL-2: Prove that n-tuples $(x_1, \cdots, x_n)$ is uniquely determined by v consist of the any basis set $\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$.

Coordinate Systems

NL-2: Prove that n-tuples $(x_1, \cdots, x_n)$ is uniquely determined by v consist of the any basis set $\{\mathbf{v}_1, \cdots, \mathbf{v}_n\}$.

(proof)

let V be a vector space. let $\mathbf{v}_1, \cdots, \mathbf{v}_n$ be linearly independent elements of V. let $x_1, \cdots, x_n$ and $y_1, \cdots, y_n$ be numbers. suppose that we have

$$x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n = y_1\mathbf{v}_1 + \cdots + y_n\mathbf{v}_n.$$

subtracting the right-hand side form the left-hand side, we get

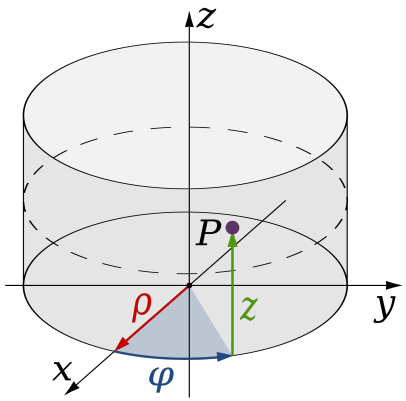
$$(x_1 - y_1)\mathbf{v}_1 + \cdots + (x_n - y_n)\mathbf{v}_n = 0.$$

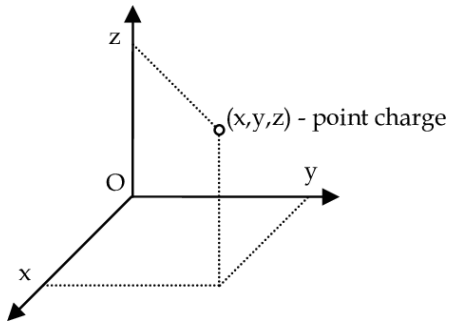
by definition, we must have $x_i - y_i = 0$ for all $i = 1, \cdots, n$, thereby proving our assertion.

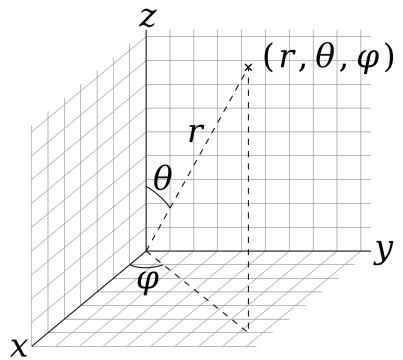
Coordinate Systems

Application to the real world.

- so far, we have algebraically explored what vector spaces, vectors, and coordinates are. by geometrically applying the fact that an n-tuple vector can be defined in a vector space based on a specific basis, we can correspond a specific point in space to a single vector.
- when you imagine the word "vector", the first thing that may come to mind is an arrow. however, the most important meaning of a vector is that it can correspond to a specific point in space as a pair of numbers in an n-tuple .
- it is also important to decide which coordinate system, that is, which basis (or basis vectors) to use. — Cartesian coordinates or spherical polar coordinates . . .







Derivative of Functions and Vectors

Mathematical representation

- derivative of a real-valued function
  - when a real-valued function $f(x)$ exists, the differentiation refers to determining the ratio of the infinitesimal change – differential in the function value with respect to an infinitesimal change in the variable, and associating it with each of the variables.

$$f'(x) = \frac{df}{dx} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x}$$

- derivative of a vector-valued function
  - A vector-valued function is a function of the form

$$\mathbf{r}(t) = f(t)\mathbf{e}_1 + g(t)\mathbf{e}_2 + h(t)\mathbf{e}_3 + \dots$$

where the component functions f, g and h, are real-valued functions of the parameter t. vector-valued functions are also written in the form

$$\mathbf{r}(t) = < f(t), g(t), h(t), \cdots > \quad \Longleftarrow \quad n - \text{tuple } X = (x_1(t), \cdots, x_n(t))$$

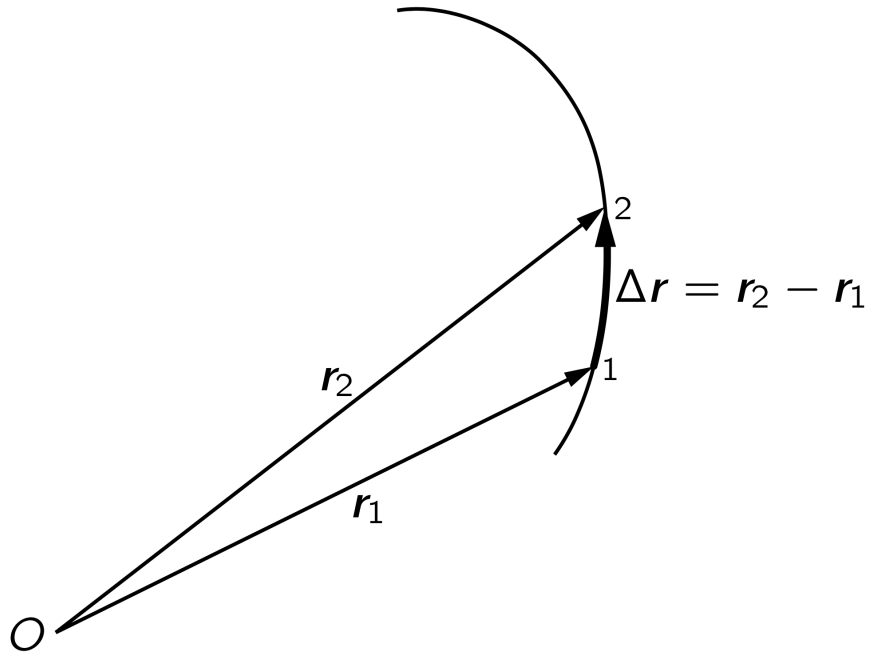
  - the derivative of a vector-valued function $\mathbf{r}(t)$ is

$$\mathbf{r}'_t = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

Position, Velocity, Acceleration

Physical representation

- position
  - » position is a vector!
  - » a point in space recorded using a coordinate system typically defined in a $\mathbb{R}^3$ Euclidean space.
  - » in physics, positions are relative to other objects or observers rather than absolute origins, since all situations are described using relative coordinate systems and relative position vector $\mathbf{r}(t)$.



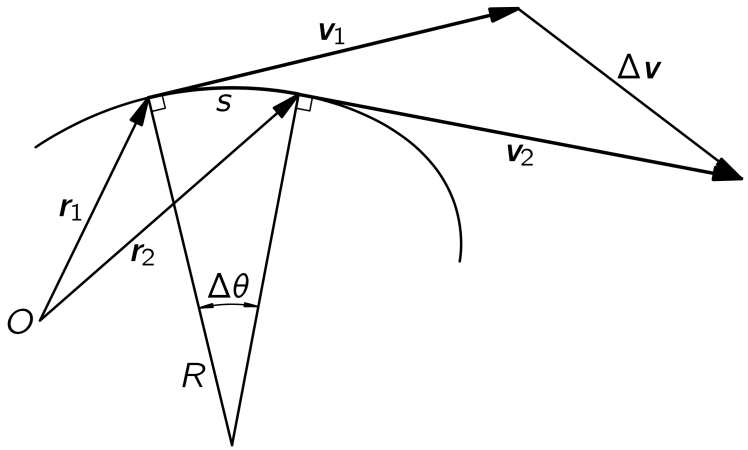
Position, Velocity, Acceleration

- velocity
  - » velocity is a vector!
  - » velocity is the directional speed of an object in motion as an indication of its rate of change in position as observed form a particular reference frame per unit time.

$$v = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

  - » or, using the position vector $\mathbf{r}(t)$

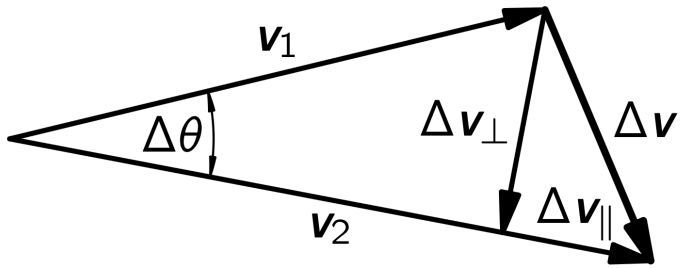
$$\mathbf{v} = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{r}}{\Delta t} = \frac{d\mathbf{r}}{dt}$$



Position, Velocity, Acceleration

- acceleration
  - » acceleration is a vector!
  - » acceleration is defined as the time rate of velocity.
  - » in the same way that the velocity i s the derivative of the distance, the acceleration is the derivative dv/dt

$$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} \qquad \text{or,} \qquad \mathbf{a} = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{v}}{\Delta t} = \frac{d\mathbf{v}}{dt}$$



NL-3: draw the parabola described by a falling body with an initial horizontal velocity.

Position, Velocity, Acceleration

NL-3: draw the parabola described by a falling ball with an initial horizontal velocity.

we shall take a motion in which a ball moves horizontally with a constant velocity u, and at the same time goes vertically downward with constant acceleration $-g$; the motion is $dx/dt = v_x = u$. since the velocity v_x is constant,

$$x = ut,$$

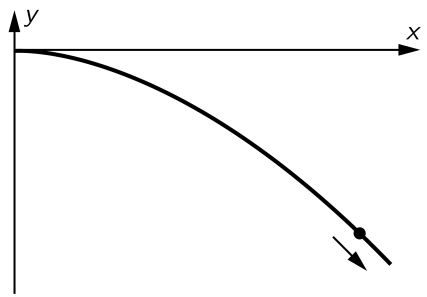
and since the downward acceleration $-g$ is constant, the distance y the object falls can be written as

$$y = -\frac{1}{2}gt^2.$$

the curve of its path can be obtained as eliminate t form equation above. since $t = t/u$. when we make this subsititution we find that

$$y = -\frac{g}{2u^2}x^2.$$

this relation between y and x may be considered as the equation of the path of the moving ball. when this equation plotted we obtain a curve that is called a parabola; any freely falling body that is shot out in any direction will travel in a parabola, as shown below.



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Point Particle and Large Bodies

what is the position of an object of any size/shape/internal structure?

1. point particle: the position is well-defined, but they do not exist in nature.
2. small bodies: effectively regarded as point particles, ignoring the size effect, which is justified only when their size should be much smaller than the distances between them.
3. large extended bodies: divided up into a large number of very small bodies (each of which may be regarded as a point particle)
   $\rightarrow$ it can be treated as a point particle located at the center of mass.

Newton's Second and Third Laws

For an isolated system of N point particles ($i = 1, 2, \ldots, N$) whose masses and positions are denoted by m_i and $\mathbf{r}_i(t)$, with respect to a given inertial frame,

Newton's second law

The equation of motion for the ith body is given by

$$\mathbf{F}_i = \frac{d}{dt}(m_i \mathbf{v}_i) = m_i \mathbf{a}_i = m_i \frac{d^2 \mathbf{r}_i}{dt^2},$$

where $\mathbf{F}_i$ is the total force acting on the body and is composed of a sum of forces due to each of the other bodies in the system:

$$\mathbf{F}_i = \sum_{j=1}^{N} \mathbf{F}_{i \leftarrow j}$$

Here $\mathbf{F}_{i \leftarrow j}$ is the force on the ith body due to the jth body and $\mathbf{F}_{i \leftarrow j} = 0$.

Newton's Second and Third Laws

Newton's third law

The two-body forces satisfy the assertion that the action and reaction are equal and opposite:

$$\mathbf{F}_{i\leftarrow j} = -\mathbf{F}_{i\leftarrow j},$$

where $\mathbf{F}_{i\leftarrow j}$ is a function of the relative position $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ and the relative velocity $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$, according to the relativity principle, and is not affected by the presence of the other bodies (i.e., there is no three-body interaction).

- perfect predictability of classical mechanics: if the forces are known and the initial positions and velocities of the bodies are given, one can solve the equations and predict the future motion of the bodies.
- there is an implicit assumption of perfect knowledge and infinite precision of calculation, which is, in general, false $\leftarrow$ chaotic theory.
- even without Newton's second law, since all fundamental forces in nature satisfy the properties of the third law, adding them together naturally satisfies the third law. In fact, it is unnecessary.

What is the Force?

$$\mathbf{F}_i = m_i \mathbf{a}_i$$

- superficial meaning
  - » if a body is accelerating, then there is a force on it. i.e., force is the mass of an object times the acceleration. but. . .
  - » Newton's second laws says nothing about the "Force". such things certainly cannot be the content of physics, because forces, masses, accelerations are definitions going in a circle.
- real content of Newton's law
  - » the force is supposed to have some independent properties, in addition to the fact that the law is described as $\mathbf{F} = m\mathbf{a}$.
  - » but the specific independent properties that the force has were not completely described by Newton or by anybody else, and therefore the physical law $\mathbf{F} = m\mathbf{a}$ is an incomplete law.
  - » can we insist upon a precise definition of force? $\Longrightarrow$ never get it!
  - » Newton's second law is not exact, and second, because in order to understand physical laws you must understand that they are all some kind of approximation.

What is the Force?

Physics

- approximation and idealization
  - because there are not any single, left-alone objects in the world — every object is a mixture of a lot of things, so we can deal with it only as a series of approximations and idealizations.
  - if we are not too precise we may idealize some object as a definite thing.
  - so we can deal with it only as a series of approximations and idealizations.

Mathematics

- difference between physics and mathematics
  - physics concerns itself with entities that are notoriously elusive to our understanding, while in mathematics, the essence of the subject of discourse is considered self-evident and requires no further explication.
  - all mathematical objects derived from mathematical laws and logic can be well-defined. however...
  - for instance wrt Euclidean geometry, whether or not those lines of Euclid, which are really abstract, apply to the lines of experience is a question for experience; it is not a question that can be answered by sheer reason.

What is the Force? (conclusion)

- conclusion for the Newton's second law
  - » we cannot just call $\mathbf{F} = m\mathbf{a}$ a definition.
  - » as dynamics is the study of describing nature, inferring all results solely based on mathematics is not true dynamics.
  - » we cannot make a mathematics of the world, because we have to find out whether the axioms are valid for the objects of nature.

Laws for Conservative Two-body Forces

- conservative force $\rightarrow$ conserved quantity — total energy of the system
- central, conservative forces between two particles

$$\mathbf{F}_{i\leftarrow j} = \mathbf{F}_{ij} = \hat{\mathbf{r}}_{ij} f(r_{ij}) \qquad \text{with } \hat{\mathbf{r}}_{ij} = \frac{\mathbf{r}_{ij}}{r_{ij}} \text{ and } r_{ij} = |\mathbf{r}_{ij}|$$

  the force is repulsive/attractive along the line joining the bodies if $f(r_{ij})$ is positive/negative.

- two laws of central, conservative forces
  - » gravitational force

$$F(r_{ij}) = -G\frac{m_i m_j}{r_{ij}^2}$$

    with the gravitational constant $G = 6.673 \times 10^{-11}\text{Nm}^2\text{kg}^{-2}$
  - » electrostatic Coulomb force

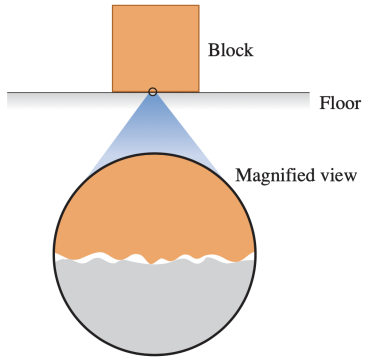
$$F(r_{ij}) = \frac{1}{4\pi\varepsilon_0}\frac{q_i q_j}{r_{ij}^2}$$

    with $1/4\pi\varepsilon_0 = 8.98755 \times 10^9\text{Nm}^2\text{C}^{-2}$

Laws for conservative Two-body Forces

- large bodies with central, conservative forces between their constituent parts
  - » conservative force between two bodies (if some further conditions are met).
  - » no longer central force; it is not directed along the line joining the bodies.

- non-conservative, velocity-dependent forces
  - » resistive and frictional forces = macroscopic effects of forces which are really conservative on a small scale
  - » transferring a part of energy from the large-scale motion of the bodies to small-scale movements in their interior $\rightarrow$ non-conservative on a large scale.
  - » if all the movements are taken into account, the total energy is conserved



$\Rightarrow$ central, conservative forces may be regarded as a norm.

Laws for conservative Two-body Forces

- the concepts of classical mechanics cannot be applied to the really really small-scale structure of matter $\rightarrow$ quantum mechanics
- electromagnetic force (Lorentz force)

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

» force between two charges in relative motion: central ($\times$), conservative ($\times$), Newton's third law ($\times$)

$\leftarrow$ finite velocity of propagation of electromagnetic waves

- the force on one charge depends not only on the instantaneous position of the other, but on its past history.
- the effect of a disturbance of one charge is felt only after an interval of time sufficient for a light signal to propagate from one to the other.

$\Rightarrow$ concept of the electromagnetic field, not force

- charges interact with fields in its immediate vicinity, not directly with the other
- field carrying energy and momentum $\rightarrow$ conservation laws

» contradiction between the relativity principle and classical electromagnetic theory

- electromagnetic theory is already consistent with Einstein's theory of special relativity in that the speed of light is constant in all the inertial frame
- according to the classical relativity principle, the speed of light can be different for relatively moving inertial frames $\rightarrow$ all inertial frames are not equivalent.
- classical mechanics and electromagnetic theory can be used together only by ignoring the relativity principle and sticking to one preferred inertial frame.

Measurement of Mass and Force

- Newton's law $\rightarrow$ new concept of mass and force
  - » any measurement of mass and force must involve the Newton's laws themselves in its interpretation
  - » Newton's law = (operational) definition of mass and force
- mass
  1. inertial mass: an object's resistance to being accelerated by a force, appearing in Newton's second law

$$\mathbf{F} = m_{\text{inertial}}\mathbf{a}$$

$\rightarrow$ an object with small inertial mass accelerates more than an object with large inertial mass when acted upon by the same force

  2. gravitational mass: it determines the strength of mutual gravitational attraction to other bodies

$$F(r_{ij}) = -G\frac{m_{i,\text{gravitational}}\, m_{j,\text{gravitational}}}{r_{ij}^2}$$

Measurement of Mass and Force

- weak equivalence principle
  - » in classical mechanics, there is not fundamental reason why two masses are equal to each other.
  - » Galileo's observations of falling bodies

$$m_{\text{inertial}}\mathbf{a} = m_{\text{gravitational}}\mathbf{g} \quad \rightarrow \quad \mathbf{a} = \frac{m_{\text{gravitational}}}{m_{\text{inertial}}}\mathbf{g}$$

    if all objects fall at the same rate in a given gravitational force
    $\rightarrow m_{\text{inertial}} \propto m_{\text{gravitational}}$
    $\rightarrow m_{\text{inertial}} = m_{\text{gravitational}}$ by choosing the mass unit appropriately
  - » two masses are same only empirically in classical mechanics

- (strong) equivalence principle
  - » in Einstein's theory of general relativity, it is impossible to distinguish between a uniform acceleration and a uniform gravitational field.
  - » the force acting on a massive object caused by a gravitational field is a result of the object's tendency to move in a straight line (in other words its inertia)
    $\rightarrow m_{\text{inertial}} = m_{\text{gravitational}}$

Measurement of Mass and Force

- measurement of inertial mass
  - » comparison of two masses = conmparison of accelerations under a same force

$$\mathbf{F} = m_1\mathbf{a}_1 = m_2\mathbf{a}_2$$

but, it does not help unless we have some way of knowing that the forces are equal (measurement of force is not defined)

  - » combining Newton's second and third laws: for two bodies mutually interacting with each other

$$m_1\mathbf{a}_1 = \mathbf{F}_{1\leftarrow 2} = -\mathbf{F}_{2\leftarrow 1} = -m_2\mathbf{a}_2 \quad \rightarrow \quad m_1\mathbf{a}_1 + m_2\mathbf{a}_2 = 0$$

if there is no influence from other remote bodies during the collision of two bodies, the total momentum is conserved

$$m_1\mathbf{v}_1 + m_2\mathbf{v}_2 = (\text{constant})$$

$\rightarrow$ measurements of velocities before/after the collision $\rightarrow$ the mass ratio

NL-4: An object A moving with velocity $\mathbf{v}$ collides with a stationary object B. After the (elastic) collision, A is moving with velocity $\frac{1}{2}\mathbf{v}$ and B with velocity $\frac{3}{2}\mathbf{v}$. Find the ratio of their masses. If, instead of boucing apart, the two bodies stuck together after the collision, with what velocity would be they then move?

Measurement of Mass and Force

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Suppose m_A and m_B are the masses of the objects A and B, respectively. Since the total momentum should be conserved in the absence of forces from outside and internal forces inside the objects after an before the collision, we get

$$m_A \times \mathbf{v} + m_B \times 0 = m_A \times \frac{1}{2}\mathbf{v} + m_B \times \frac{3}{2}\mathbf{v}, \quad \text{or} \quad m_A = 3m_B.$$

Therefore, the ratio of two masses is $m_A/m_B = 3$.
In the second case, the conservation of the total momentum gives rise to

$$m_A \times \mathbf{v} + m_B \times 0 = (m_A + m_B)\mathbf{v}_f,$$

where $\mathbf{v}_f$ is the final velocity of the composite body. By substituting m_A with $3m_B$, one gets

$$\mathbf{v}_f = \frac{3}{4}\mathbf{v}.$$

Measurement of Mass and Force

A1: Axiom for acceleration relation of two bodies

In an isolated two-body system, the acceleration always satisfy the relation $\mathbf{a}_1 = -k_{21}\mathbf{a}_2$, where the scale k_{21} is, for two given bodies, a constant independent of their positions, velocities, and internal states.

Corollary of A1

$$k_{21} = \frac{1}{k_{12}}, k_{31} = k_{32}k_{21}$$

A2: Axiom for acceleration in many-body cases

The acceleration induced in one body by another is some definite function of their positions, velocities, and internal structure, and is unaffected by the presence of other bodies. In a many-body system, the acceleration of any given body is equal to the sum of the accelerations induced in it by each of the other bodies individually.

- Newton's second/third laws = A1 + A2 (without the concept of the mass and force)

Measurement of Mass and Force

- using the concept of “mass” and “force” is better since
  - » the action-backaction relation is expressed in a simpler way by using the force: $\mathbf{F}_{12} = -\mathbf{F}_{21}$ *vs* $\mathbf{a}_2 = k_{21}\mathbf{a}_1$
  - » allocating a mass for each object is much better than finding out the ratio $k_{21} = m_2/m_1$ for each pair of two bodies.
  - » necessarily, the acceleration of one body should be dependent on its mass, demanding the introduction of the concept of the mass.
- additive nature of the mass: the mass of the two bodies rigidly bound together is imply the sum of two masses

$$m_{1+2} = m_1 + m_2 \tag{1}$$

NL-5: proof of Eq. (1)

Measurement of Mass and Force

NL-5: proof of Eq. (1)

(proof)

Let us take a three-body system. The equations of motion for the three bodies then are

$$m_1\mathbf{a}_1 = \mathbf{F}_{12} + \mathbf{F}_{13}$$
$$m_2\mathbf{a}_2 = \mathbf{F}_{21} + \mathbf{F}_{23}$$
$$m_3\mathbf{a}_3 = \mathbf{F}_{31} + \mathbf{F}_{32}$$

By adding all these equations and applying the Newton's third law,

$$m_1\mathbf{a}_1 + m_2\mathbf{a}_2 + m_3\mathbf{a}_3 = (\mathbf{F}_{12} + \mathbf{F}_{23}) + (\mathbf{F}_{23} + \mathbf{F}_{32}) + (\mathbf{F}_{31} + \mathbf{F}_{13}) = 0$$

Now suppose that the force between the second and third body is such that they are rigidly bound together to form a composite body. Then their acceleration must be equal

$$\mathbf{a}_2 = \mathbf{a}_3 \equiv \mathbf{a}_{2+3}$$

which shows that the mass of the composite body is just $m_{2+3} = m_2 + m_3$.

External Forces

- many-body problem: equation of motion for interacting two bodies

$$m_1\mathbf{a}_1 = \mathbf{F}_{12}(\mathbf{r}_1 - \mathbf{r}_2, \mathbf{v}_1 - \mathbf{v}_2)$$
$$m_2\mathbf{a}_2 = \mathbf{F}_{21}(\mathbf{r}_1 - \mathbf{r}_2, \mathbf{v}_1 - \mathbf{v}_2)$$

  - » positions, velocities at a time $\rightarrow$ forces acting on the bodies
    $\rightarrow$ new positions, velocities after a short interval $\rightarrow$ new forces $\rightarrow \ldots$

  $\Rightarrow$ the whole equations should be solved simultaneously in a self-consistent manner, which is, in many cases, very difficult

- approximation to single-particle problem
  - » suppose that we are interested in a small part of a dynamical system and the other part is much larger so that the effect of the small part on the other part can be neglected.
  - » example: case of Earth and artificial satellite
  - » external forces: suppose that $m_2 \gg m_1$

$$m_2\mathbf{a}_2 = \mathbf{F}_{21}(\mathbf{r}_1 - \mathbf{r}_2, \mathbf{v}_1 - \mathbf{v}_2) \approx 0$$
$$m_1\mathbf{a}_1 \approx \mathbf{F}_{\text{ext}}(\mathbf{r}_1, \mathbf{v}_1)$$

  the forces due to the "environment" are already known in advance (and do not change due to the motion of the small system of our interest) can be regarded as external forces $\rightarrow$ non-interacting single-particle problem.

Fields: Fundamental Forces

- when particles are at rest, their force follows a simple inverse-square relationship with distance. but as particles move, the force between them becomes highly complex.
  $\longrightarrow$ leading to the introduction of the concept of a field for mathematical description.

Electric Field

$$\mathbf{F}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\,\hat{\mathbf{r}}$$

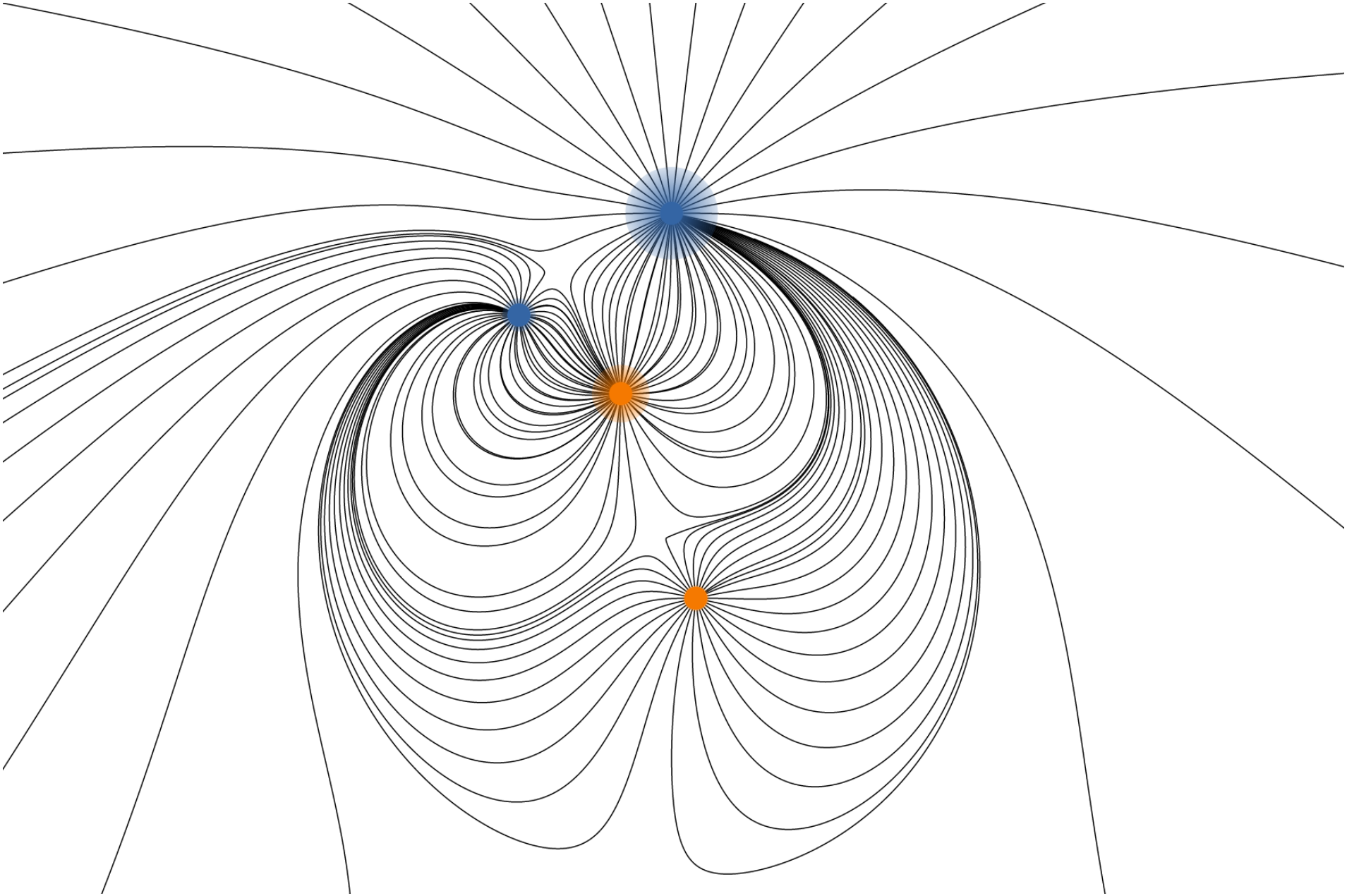
- the charge q_1 located at P produces a condition at R, such that when the charge q_2 is placed at R feels the force. it is q_2 multiplied by a quantity $\mathbf{E}$ that would be there whether q_2 were there or not.
- $\mathbf{E}$ is the condition produced by q_1, we say, and $\mathbf{F}$ is the response of q_2 to $\mathbf{E}$. then $\mathbf{E}$ is called an electric field.

$$\mathbf{E} = \frac{q_1 \mathbf{r}}{4\pi\varepsilon_0 r^3}$$

we then write

$$\mathbf{F} = q_2\mathbf{E}$$

Fields: Fundamental Forces



Fields: Fundamental Forces

- what is the point of all this?
  $\longrightarrow$ dividing the analysis into two parts!

$$\mathbf{F}(\mathbf{r}) \quad \Longleftrightarrow \quad q_2\mathbf{E}(\mathbf{r})$$

» if the laws of force were simple, there would be no need to introduce the concept. However, since force operates in a highly complex manner, separating it into "fields created at a point" and "physical quantities indicating the degree of sensitivity to the field".

» it turns out that the fields have a reality that is almost independent of the objects which create them.

» to determine the force applied to a particle, you need to know the history of all other particles because there is a time delay in the transmission of the force between particles. this is why the force operates in a highly complex manner.

Exercises

NL-6: A neutron star is a collection of neutrons bound together by their mutual gravitation with a density comparable to that of an atomic nucleus (approximately $10^{12}\mathrm{g}/\mathrm{cm}^3$. Assume that the neutron star is a sphere with a uniform density ρ and show that the maximum frequency with which it may rotate, if mass if not to fly off at the equator, is $f = (\rho G/3\pi)^{1/2}$. Calculate f for a density of $10^{12}\mathrm{g}/\mathrm{cm}^3$). It has been suggested that pulsars, which emit regular bursts of radiation at repetition rates up to about $30/\mathrm{sec}$, are rotating neutron stars.

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Suppose that R is the radius of the neutron star. Consider a neutron (of mass m) at the equator. In order for the neutron not to fly off, the gravitational attraction should be greater that or equal to the required centripetal force:

$$G\frac{Mm}{R^2} \geq mR(2\pi f)^2$$

where $m = \rho \times \frac{4}{3}\pi R^3$ is the total mass of the neutron star. Then, we get

$$f \leq \frac{1}{2\pi}\sqrt{G\frac{M}{R^3}} = \sqrt{G\rho\frac{4}{3}\frac{\pi R^3}{4\pi^2 R^3}} = \sqrt{\frac{\rho G}{3\pi}} \equiv f_{\mathrm{max}}$$

For $\rho = 10^{12}\mathrm{g/cm}^3$,

$$f_{\mathrm{max}} = 84/\mathrm{sec}$$